2.43 2nd Order DEs: Non-Homogeneous

Worked Solutions

1 a)

$$
\begin{aligned}
y &= Ae^{-2x} + Be^{-3x} \\
\frac{dy}{dx} &= -2Ae^{-2x} - 3Be^{-3x} \\
\frac{d^2y}{dx^2} &= 4Ae^{-2x} + 9Be^{-3x} \\
\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y &= 0 \\
\text{LHS: } & 4Ae^{-2x} + 9Be^{-3x} \\
& + 5(-2Ae^{-2x} - 3Be^{-3x}) \\
& + 6(Ae^{-2x} + Be^{-3x}) \\
& = 4Ae^{-2x} + 9Be^{-3x} - 10Ae^{-2x} - 15Be^{-3x} + 6Ae^{-2x} + 6Be^{-3x} \\
& = (4 - 10 + 6)Ae^{-2x} + (9 - 15 + 6)Be^{-3x} \\
& = 0 : \text{RHS.}
\end{aligned}
$$

1 b)

$$
\begin{aligned}
y &= k \\
\frac{dy}{dx} &= 0 \\
\frac{d^2y}{dx^2} &= 0 \\
\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y &= 0 + 5 \times 0 + 6 \times k \\
&= 6k \text{ as required.}
\end{aligned}
$$

1 c)

$$
\begin{aligned}
&6K = 3 \Rightarrow K = \frac{1}{2} \\
&\therefore y = \frac{1}{2} \text{ is a solution of the differential equation.}
\end{aligned}
$$

1 d)

$$
\begin{aligned}
y &= Ae^{-2x} + Be^{-3x} + \frac{1}{2} \\
\frac{dy}{dx} &= -2Ae^{-2x} - 3Be^{-3x} \\
\frac{d^2y}{dx^2} &= 4Ae^{-2x} + 9Be^{-3x}
\end{aligned}
$$

$$\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 0$$

LHS: $4Ae^{-2x} + 9Be^{-3x}$

$+ 5(-2Ae^{-2x} - 3Be^{-3x})$

$+ 6(Ae^{-2x} + Be^{-3x} + \frac{1}{2})$

$= 4Ae^{-2x} + 9Be^{-3x} - 10Ae^{-2x} - 15Be^{-3x} + 6Ae^{-2x} + 6Be^{-3x} + 6$

$= (4 - 10 + 6)Ae^{-2x} + (9 - 15 + 6)Be^{-3x} + 6$

$= 0 + 0 + 6$

$= 6$: RHS.

2 a)

$$\begin{aligned} y &= k \\ \frac{dy}{dx} &= 0 \\ \frac{d^2y}{dx^2} &= 0 \\ \frac{d^2y}{dx^2} - \frac{dy}{dx} - 20y &= 5 \\ \Rightarrow 0 - 0 - 20 \times k &= 5 \\ \Rightarrow k &= -\frac{1}{4} \end{aligned}$$

Particular Integral: $y = -\frac{1}{4}$.

2 b)

$$\begin{aligned} y &= k \\ \frac{dy}{dx} &= 0 \\ \frac{d^2y}{dx^2} &= 0 \\ 4\frac{d^2y}{dx^2} - 9y &= 6 \\ \Rightarrow 4 \times 0 - 9 \times k &= 6 \\ \Rightarrow k &= -\frac{2}{3} \end{aligned}$$

Particular Integral: $y = -\frac{2}{3}$.

2 c)

$$\begin{aligned} y &= k \\ \frac{dy}{dx} &= 0 \end{aligned}$$

$$\frac{d^2y}{dx^2} = 0$$
$$4\frac{d^2y}{dx^2} + 11\frac{dy}{dx} - 3y = -60$$
$$\Rightarrow 4 \times 0 + 11 \times 0 - 3 \times k = -60$$
$$\Rightarrow k = 20$$
$$\text{Particular Integral: } y = 20.$$

3 a)

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} - 20y = 5$$
$$\text{Homogeneous Equation: } \frac{d^2y}{dx^2} - \frac{dy}{dx} - 20y = 0$$
$$\text{Auxiliary Equation: } m^2 - m - 20 = 0$$
$$\Rightarrow (m-5)(m+4) = 0$$
$$\Rightarrow m = 5 \text{ or } m = -4$$
$$\text{Complementary Function: } y = Ae^{5x} + Be^{-4x}$$
$$\text{Particular Integral: } y = -\frac{1}{4}$$
$$\text{General Solution: } y = Ae^{5x} + Be^{-4x} - \frac{1}{4}.$$

3 b)

$$4\frac{d^2y}{dx^2} - 9y = 6$$
$$\text{Homogeneous Equation: } 4\frac{d^2y}{dx^2} - 9y = 0$$
$$\text{Auxiliary Equation: } 4m^2 - 9 = 0$$
$$\Rightarrow m^2 = \frac{9}{4}$$
$$\Rightarrow m = \pm\frac{3}{2}$$
$$\text{Complementary Function: } y = Ae^{\frac{3}{2}x} + Be^{-\frac{3}{2}x}$$
$$\text{Particular Integral: } y = -\frac{2}{3}$$
$$\text{General Solution: } y = Ae^{\frac{3}{2}x} + Be^{-\frac{3}{2}x} - \frac{2}{3}.$$

3 c)

$$4\frac{d^2y}{dx^2} + 11\frac{dy}{dx} - 3y = -60$$
$$\text{Homogeneous Equation: } 4\frac{d^2y}{dx^2} + 11\frac{dy}{dx} - 3y = 0$$
$$\text{Auxiliary Equation: } 4m^2 + 11m - 3 = 0$$

$$\Rightarrow (4m-1)(m+3) = 0$$
$$\Rightarrow m = \frac{1}{4} \text{ or } m = -3$$
$$\text{Complementary Function: } y = Ae^{\frac{1}{4}x} + Be^{-3x}$$
$$\text{Particular Integral: } y = 20$$
$$\text{General Solution: } y = Ae^{\frac{1}{4}x} + Be^{-3x} + 20.$$

4 a)

$$y = k$$
$$\frac{dy}{dx} = 0$$
$$\frac{d^2y}{dx^2} = 0$$
$$\frac{d^2y}{dx^2} - \frac{dy}{dx} = 8$$
$$\Rightarrow 0 - 0 = 8$$
$$\text{As } 0 \neq 8, \text{ there is no value of } k \text{ that satisfies the equation.}$$

4 b)

$$y = ax + b$$
$$\frac{dy}{dx} = a$$
$$\frac{d^2y}{dx^2} = 0$$
$$\frac{d^2y}{dx^2} - \frac{dy}{dx} = 8$$
$$\Rightarrow 0 - a = 8$$
$$\Rightarrow a = -8$$
$$\text{The constant } b \text{ can take on any value.}$$
$$\text{Particular Integral: } y = -8x + b.$$

4 c)

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} = 8$$
$$\text{Homogeneous Equation: } \frac{d^2y}{dx^2} - \frac{dy}{dx} = 0$$
$$\text{Auxiliary Equation: } m^2 - m = 0$$
$$\Rightarrow m(m-1) = 0$$
$$\Rightarrow m = 0 \text{ or } m = 1$$
$$\text{Complementary Function: } y = Ae^0 + Be^x$$
$$\Rightarrow y = A + Be^x$$

$$\text{Particular Integral: } y = -8x + b$$
$$\text{General Solution: } y = A + Be^x - 8x + b$$
$$\Rightarrow y = C + Be^x - 8x$$
$$\text{where } C = A + b.$$

5)

$$y = ax + b$$
$$\frac{dy}{dx} = a$$
$$\frac{d^2y}{dx^2} = 0$$
$$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} - 10y = 10x + 7$$
$$\Rightarrow 0 + 3a - 10(ax + b) = 10x + 7$$
$$\Rightarrow (-10a)x + (3a - 10b) = 10x + 7$$
$$\text{Compare coefficients: } -10a = 10 \Rightarrow a = -1$$
$$3(-1) - 10b = 7 \Rightarrow -10b = 10 \Rightarrow b = -1$$
$$\text{Particular Integral: } y = -x - 1$$
$$\therefore a = -1 \text{ and } b = -1.$$

6 a)

$$y = ax + b, \quad \frac{dy}{dx} = a, \quad \frac{d^2y}{dx^2} = 0$$
$$\frac{d^2y}{dx^2} + 7\frac{dy}{dx} - 30y = 15x + 4$$
$$\Rightarrow 0 + 7a - 30(ax + b) = 15x + 4$$
$$\Rightarrow (-30a)x + (7a - 30b) = 15x + 4$$
$$\text{Compare coefficients: } -30a = 15 \Rightarrow a = -\frac{1}{2}$$
$$7(-\frac{1}{2}) - 30b = 4 \Rightarrow b = -\frac{1}{4}$$
$$\text{Particular Integral: } y = -\frac{1}{2}x - \frac{1}{4}.$$

6 b)

$$y = ax + b, \quad \frac{dy}{dx} = a, \quad \frac{d^2y}{dx^2} = 0$$
$$9\frac{d^2y}{dx^2} - y = 5 - x$$
$$\Rightarrow 0 - (ax + b) = 5 - x$$
$$\Rightarrow (-a)x - b = -x + 5$$
$$\text{Compare coefficients: } -a = -1 \Rightarrow a = 1$$

$$-b = 5 \Rightarrow b = -5$$

Particular Integral: $y = x - 5$.

6 c)

$$y = ax + b, \quad \frac{dy}{dx} = a, \quad \frac{d^2y}{dx^2} = 0$$

$$4\frac{d^2y}{dx^2} + 16\frac{dy}{dx} - 9y = -45x$$

$$\Rightarrow 0 + 16a - 9(ax + b) = -45x$$

$$\Rightarrow (-9a)x + (16a - 9b) = -45x$$

Compare coefficients: $-9a = -45 \Rightarrow a = 5$

$$16(5) - 9b = 0 \Rightarrow b = \frac{80}{9}$$

Particular Integral: $y = 5x + \frac{80}{9}$.

7 a)

$$\frac{d^2y}{dx^2} + 7\frac{dy}{dx} - 30y = 15x + 4$$

Homogeneous Equation: $\frac{d^2y}{dx^2} + 7\frac{dy}{dx} - 30y = 0$

Auxiliary Equation: $m^2 + 7m - 30 = 0$

$$\Rightarrow (m + 10)(m - 3) = 0$$

$$\Rightarrow m = -10 \text{ or } m = 3$$

Complementary Function: $y = Ae^{-10x} + Be^{3x}$

Particular Integral: $y = -\frac{1}{2}x - \frac{1}{4}$

General Solution: $y = Ae^{-10x} + Be^{3x} - \frac{1}{2}x - \frac{1}{4}$.

7 b)

$$9\frac{d^2y}{dx^2} - y = 5 - x$$

Homogeneous Equation: $9\frac{d^2y}{dx^2} - y = 0$

Auxiliary Equation: $9m^2 - 1 = 0$

$$\Rightarrow m^2 = \frac{1}{9}$$

$$\Rightarrow m = \pm\frac{1}{3}$$

Complementary Function: $y = Ae^{\frac{1}{3}x} + Be^{-\frac{1}{3}x}$

Particular Integral: $y = x - 5$

General Solution: $y = Ae^{\frac{1}{3}x} + Be^{-\frac{1}{3}x} + x - 5$.

7 c)

$$4\frac{d^2y}{dx^2} + 16\frac{dy}{dx} - 9y = -45x$$

Homogeneous Equation: $4\frac{d^2y}{dx^2} + 16\frac{dy}{dx} - 9y = 0$

Auxiliary Equation: $4m^2 + 16m - 9 = 0$

$$\Rightarrow (2m + 9)(2m - 1) = 0$$

$$\Rightarrow m = -\frac{9}{2} \text{ or } m = \frac{1}{2}$$

Complementary Function: $y = Ae^{-\frac{9}{2}x} + Be^{\frac{1}{2}x}$

Particular Integral: $y = 5x + \frac{80}{9}$

General Solution: $y = Ae^{-\frac{9}{2}x} + Be^{\frac{1}{2}x} + 5x + \frac{80}{9}$.

8 a)

$$y = ax + b$$

$$\frac{dy}{dx} = a$$

$$\frac{d^2y}{dx^2} = 0$$

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} = 8x$$

$$\Rightarrow 0 + 4a = 8x$$

As the left hand side has no x but the right hand side does.

There is no value of a that satisfies the equation.

8 b)

$$y = ax^2 + bx + c$$

$$\frac{dy}{dx} = 2ax + b$$

$$\frac{d^2y}{dx^2} = 2a$$

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} = 8x$$

$$\Rightarrow 2a + 4(2ax + b) = 8x$$

$$\Rightarrow (8a)x + (2a + 4b) = 8x$$

Compare Coefficients: $8a = 8 \Rightarrow a = 1$

$$2(1) + 4b = 0 \Rightarrow b = -\frac{1}{2}$$

The constant c can take on any value.

Particular Integral: $y = x^2 - \frac{1}{2}x + c$.

8 c)

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} = 8x$$

$$\text{Homogeneous Equation: } \frac{d^2y}{dx^2} + 4\frac{dy}{dx} = 0$$

$$\text{Auxiliary Equation: } m^2 + 4m = 0$$

$$\Rightarrow m(m+4) = 0$$

$$\Rightarrow m = 0 \text{ or } m = -4$$

$$\text{Complementary Function: } y = Ae^0 + Be^{-4x}$$

$$\Rightarrow y = A + Be^{-4x}$$

$$\text{Particular Integral: } y = x^2 - \frac{1}{2}x + c$$

$$\text{General Solution: } y = A + Be^{-4x} + x^2 - \frac{1}{2}x + c$$

$$\Rightarrow y = C + Be^{-4x} + x^2 - \frac{1}{2}x$$

$$\text{where } C = A + c.$$

9)

$$y = ax^2 + bx + c$$

$$\frac{dy}{dx} = 2ax + b$$

$$\frac{d^2y}{dx^2} = 2a$$

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} - 6y = 12x^2 + 6x - 3$$

$$\Rightarrow 2a - (2ax + b) - 6(ax^2 + bx + c) = 12x^2 + 6x - 3$$

$$\Rightarrow (-6a)x^2 + (-2a - 6b)x + (2a - b - 6c) = 12x^2 + 6x - 3$$

$$\text{Compare coefficients: } -6a = 12 \Rightarrow a = -2$$

$$-2(-2) - 6b = 6 \Rightarrow b = -\frac{1}{3}$$

$$2(-2) - (-\frac{1}{3}) - 6c = -3 \Rightarrow c = -\frac{1}{9}$$

$$\text{Particular Integral: } y = -2x^2 - \frac{1}{3}x - \frac{1}{9}$$

$$\therefore a = -2,\ b = -\frac{1}{3} \text{ and } c = -\frac{1}{9}.$$

10 a)

$$y = ax^2 + bx + c, \quad \frac{dy}{dx} = 2ax + b, \quad \frac{d^2y}{dx^2} = 2a$$

$$\frac{d^2y}{dx^2} - 8\frac{dy}{dx} - 9y = 18x^2 + 11x - 14$$

$$\Rightarrow 2a - 8(2ax + b) - 9(ax^2 + bx + c) = 18x^2 + 11x - 14$$

$$\Rightarrow (-9a)x^2 + (-16a - 9b)x + (2a - 8b - 9c) = 18x^2 + 11x - 14$$

$$\text{Compare coefficients: } -9a = 18 \Rightarrow a = -2$$

$$-16(-2) - 9b = 11 \Rightarrow b = \frac{7}{3}$$

$$2(-2) - 8(\frac{7}{3}) - 9c = -14 \Rightarrow c = -\frac{26}{27}$$

$$\text{Particular Integral: } y = -2x^2 + \frac{7}{3}x - \frac{26}{27}.$$

10 b)

$$y = ax^2 + bx + c, \quad \frac{dy}{dx} = 2ax + b, \quad \frac{d^2y}{dx^2} = 2a$$

$$4\frac{d^2y}{dx^2} - 16y = -8x^2$$

$$\Rightarrow 4(2a) - 16(ax^2 + bx + c) = -8x^2$$

$$\Rightarrow (-16a)x^2 + (-16b)x + (8a - 16c) = -8x^2$$

$$\text{Compare coefficients: } -16a = -8 \Rightarrow a = \frac{1}{2}$$

$$-16b = 0 \Rightarrow b = 0$$

$$8(\frac{1}{2}) - 16c = 0 \Rightarrow c = \frac{1}{4}$$

$$\text{Particular Integral: } y = \frac{1}{2}x^2 + \frac{1}{4}.$$

10 c)

$$y = ax^2 + bx + c, \quad \frac{dy}{dx} = 2ax + b, \quad \frac{d^2y}{dx^2} = 2a$$

$$9\frac{d^2y}{dx^2} - 9\frac{dy}{dx} + 2y = 11 - 4x - 10x^2$$

$$\Rightarrow 9(2a) - 9(2ax + b) + 2(ax^2 + bx + c) = 11 - 4x - 10x^2$$

$$\Rightarrow (2a)x^2 + (-18a + 2b)x + (18a - 9b + 2c) = -10x^2 - 4x + 11$$

$$\text{Compare coefficients: } 2a = -10 \Rightarrow a = -5$$

$$-18(-5) + 2b = -4 \Rightarrow b = -47$$

$$18(-5) - 9(-47) + 2c = 11 \Rightarrow c = -161$$

$$\text{Particular Integral: } y = -5x^2 - 47x - 161.$$

11 a)

$$\frac{d^2y}{dx^2} - 8\frac{dy}{dx} - 9y = 18x^2 + 11x - 14$$

$$\text{Homogeneous Equation: } \frac{d^2y}{dx^2} - 8\frac{dy}{dx} - 9y = 0$$

$$\text{Auxiliary Equation: } m^2 - 8m - 9 = 0$$

$$\Rightarrow (m - 9)(m + 1) = 0$$

$$\Rightarrow m = 9 \text{ or } m = -1$$

$$\text{Complementary Function: } y = Ae^{9x} + Be^{-x}$$

$$\text{Particular Integral: } y = -2x^2 + \frac{7}{3}x - \frac{26}{27}$$

$$\text{General Solution: } y = Ae^{9x} + Be^{-x} - 2x^2 + \frac{7}{3}x - \frac{26}{27}.$$

11 b)

$$4\frac{d^2y}{dx^2} - 16y = -8x^2$$

$$\text{Homogeneous Equation: } 4\frac{d^2y}{dx^2} - 16y = 0$$

$$\text{Auxiliary Equation: } 4m^2 - 16 = 0$$

$$\Rightarrow m^2 - 4 = 0$$

$$\Rightarrow (m+2)(m-2) = 0$$

$$\Rightarrow m = -2 \text{ or } m = 2$$

$$\text{Complementary Function: } y = Ae^{-2x} + Be^{2x}$$

$$\text{Particular Integral: } y = \frac{1}{2}x^2 + \frac{1}{4}$$

$$\text{General Solution: } y = Ae^{-2x} + Be^{2x} + \frac{1}{2}x^2 + \frac{1}{4}.$$

11 c)

$$9\frac{d^2y}{dx^2} - 9\frac{dy}{dx} + 2y = 11 - 4x - 10x^2$$

$$\text{Homogeneous Equation: } 9\frac{d^2y}{dx^2} - 9\frac{dy}{dx} + 2y = 0$$

$$\text{Auxiliary Equation: } 9m^2 - 9m + 2 = 0$$

$$\Rightarrow (3m-2)(3m-1) = 0$$

$$\Rightarrow m = \frac{2}{3} \text{ or } m = \frac{1}{3}$$

$$\text{Complementary Function: } y = Ae^{\frac{2}{3}x} + Be^{\frac{1}{3}x}$$

$$\text{Particular Integral: } y = -5x^2 - 47x - 161$$

$$\text{General Solution: } y = Ae^{\frac{2}{3}x} + Be^{\frac{1}{3}x} - 5x^2 - 47x - 161.$$

12 a)

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 5y = 4x + 1$$

$$\text{Homogeneous Equation: } \frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 5y = 0$$

$$\text{Auxiliary Equation: } m^2 + 2m + 5 = 0$$

$$\Rightarrow (m+1)^2 - 1 + 5 = 0$$

$$\Rightarrow (m+1)^2 = -4$$
$$\Rightarrow m = -1 \pm 2i$$
$$\text{Complementary Function: } y = e^{-x}(A\cos 2x + B\sin 2x)$$
$$\text{Try this Particular Integral: } y = ax + b, \quad y' = a, \quad y'' = 0$$
$$\Rightarrow 0 + 2a + 5(ax + b) = 4x + 1$$
$$\Rightarrow (5a)x + (2a + 5b) = 4x + 1$$
$$\text{Compare coefficients: } 5a = 4 \Rightarrow a = \frac{4}{5}$$
$$2(\frac{4}{5}) + 5b = 1 \Rightarrow b = -\frac{3}{25}$$
$$\text{Particular Integral: } y = \frac{4}{5}x - \frac{3}{25}$$
$$\text{General Solution: } y = e^{-x}(A\cos 2x + B\sin 2x) + \frac{4}{5}x - \frac{3}{25}.$$

12 b)

$$\ddot{x} - 4\dot{x} + 4x = 4t^2 + 8t + 6$$
$$\text{Homogeneous Equation: } \ddot{x} - 4\dot{x} + 4x = 0$$
$$\text{Auxiliary Equation: } m^2 - 4m + 4 = 0$$
$$\Rightarrow (m-2)^2 = 0$$
$$\Rightarrow m = 2$$
$$\text{Complementary Function: } x = Ate^{2t} + Be^{2t}$$
$$\text{Try this Particular Integral: } x = at^2 + bt + c, \quad \dot{x} = 2at + b, \quad \ddot{x} = 2a$$
$$\Rightarrow 2a - 4(2at + b) + 4(at^2 + bt + c) = 4t^2 + 8t + 6$$
$$\Rightarrow (4a)t^2 + (-8a + 4b)t + (2a - 4b + 4c) = 4t^2 + 8t + 6$$
$$\text{Compare coefficients: } 4a = 4 \Rightarrow a = 1$$
$$-8(1) + 4b = 8 \Rightarrow b = 4$$
$$2(1) - 4(4) + 4c = 6 \Rightarrow c = 5$$
$$\text{Particular Integral: } x = t^2 + 4t + 5$$
$$\text{General Solution: } x = Ate^{2t} + Be^{2t} + t^2 + 4t + 5.$$

12 c)

$$\frac{d^2y}{dx^2} + 9y = 9x$$
$$\text{Homogeneous Equation: } \frac{d^2y}{dx^2} + 9y = 0$$
$$\text{Auxiliary Equation: } m^2 + 9 = 0$$
$$\Rightarrow m^2 = -9$$
$$\Rightarrow m = \pm 3i$$
$$\text{Complementary Function: } y = A\cos 3x + B\sin 3x$$
$$\text{Try this Particular Integral: } y = ax + b, \quad y' = a, \quad y'' = 0$$
$$\Rightarrow 0 + 9(ax + b) = 9x$$

$$\Rightarrow (9a)x + (9b) = 9x$$

$$\text{Compare coefficients: } 9a = 9 \Rightarrow a = 1$$

$$9b = 0 \Rightarrow b = 0$$

$$\text{Particular Integral: } y = x$$

$$\text{General Solution: } y = A\cos 3x + B\sin 3x + x.$$

12 d)

$$\ddot{x} - 5\dot{x} + 6x = 12$$

$$\text{Homogeneous Equation: } \ddot{x} - 5\dot{x} + 6x = 0$$

$$\text{Auxiliary Equation: } m^2 - 5m + 6 = 0$$

$$\Rightarrow (m-2)(m-3) = 0$$

$$\Rightarrow m = 2 \text{ or } m = 3$$

$$\text{Complementary Function: } x = Ae^{2t} + Be^{3t}$$

$$\text{Try this Particular Integral: } x = k, \quad \dot{x} = 0, \quad \ddot{x} = 0$$

$$\Rightarrow 0 - 5(0) + 6k = 12 \Rightarrow k = 2$$

$$\text{Particular Integral: } x = 2$$

$$\text{General Solution: } x = Ae^{2t} + Be^{3t} + 2.$$

12 e)

$$\frac{d^2y}{dx^2} + 4y = 8x^2$$

$$\text{Homogeneous Equation: } \frac{d^2y}{dx^2} + 4y = 0$$

$$\text{Auxiliary Equation: } m^2 + 4 = 0$$

$$\Rightarrow m^2 = -4$$

$$\Rightarrow m = \pm 2i$$

$$\text{Complementary Function: } y = A\cos 2x + B\sin 2x$$

$$\text{Try this Particular Integral: } y = ax^2 + bx + c, \quad y' = 2ax + b, \quad y'' = 2a$$

$$\Rightarrow 2a + 4(ax^2 + bx + c) = 8x^2$$

$$\Rightarrow (4a)x^2 + (4b)x + (2a + 4c) = 8x^2$$

$$\text{Compare coefficients: } 4a = 8 \Rightarrow a = 2$$

$$4b = 0 \Rightarrow b = 0$$

$$2(2) + 4c = 0 \Rightarrow c = -1$$

$$\text{Particular Integral: } y = 2x^2 - 1$$

$$\text{General Solution: } y = A\cos 2x + B\sin 2x + 2x^2 - 1.$$

12 f)

$$\ddot{x} + 4\dot{x} + 8x = 2t^2 + 4t$$

$$\text{Homogeneous Equation: } \ddot{x} + 4\dot{x} + 8x = 0$$
$$\text{Auxiliary Equation: } m^2 + 4m + 8 = 0$$
$$\Rightarrow (m+2)^2 - 4 + 8 = 0$$
$$\Rightarrow (m+2)^2 = -4$$
$$\Rightarrow m = -2 \pm 2i$$
$$\text{Complementary Function: } x = e^{-2t}(A\cos 2t + B\sin 2t)$$
$$\text{Try this Particular Integral: } x = at^2 + bt + c, \quad \dot{x} = 2at + b, \quad \ddot{x} = 2a$$
$$\Rightarrow 2a + 4(2at + b) + 8(at^2 + bt + c) = 2t^2 + 4t$$
$$\Rightarrow (8a)t^2 + (8a + 8b)t + (2a + 4b + 8c) = 2t^2 + 4t$$
$$\text{Compare coefficients: } 8a = 2 \Rightarrow a = \frac{1}{4}$$
$$8(\frac{1}{4}) + 8b = 4 \Rightarrow b = \frac{1}{4}$$
$$2(\frac{1}{4}) + 4(\frac{1}{4}) + 8c = 0 \Rightarrow c = -\frac{3}{16}$$
$$\text{Particular Integral: } x = \frac{1}{4}t^2 + \frac{1}{4}t - \frac{3}{16}$$
$$\text{General Solution: } x = e^{-2t}(A\cos 2t + B\sin 2t) + \frac{1}{4}t^2 + \frac{1}{4}t - \frac{3}{16}.$$

12 g)

$$\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 4x + 2$$
$$\text{Homogeneous Equation: } \frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 0$$
$$\text{Auxiliary Equation: } m^2 - 3m + 2 = 0$$
$$\Rightarrow (m-1)(m-2) = 0$$
$$\Rightarrow m = 1 \text{ or } m = 2$$
$$\text{Complementary Function: } y = Ae^x + Be^{2x}$$
$$\text{Try this Particular Integral: } y = ax + b, \quad y' = a, \quad y'' = 0$$
$$\Rightarrow 0 - 3a + 2(ax + b) = 4x + 2$$
$$\Rightarrow (2a)x + (-3a + 2b) = 4x + 2$$
$$\text{Compare coefficients: } 2a = 4 \Rightarrow a = 2$$
$$-3(2) + 2b = 2 \Rightarrow b = 4$$
$$\text{Particular Integral: } y = 2x + 4$$
$$\text{General Solution: } y = Ae^x + Be^{2x} + 2x + 4.$$

12 h)

$$25\ddot{x} + 10\dot{x} + x = 6 - t^2$$
$$\text{Homogeneous Equation: } 25\ddot{x} + 10\dot{x} + x = 0$$
$$\text{Auxiliary Equation: } 25m^2 + 10m + 1 = 0$$
$$\Rightarrow (5m+1)^2 = 0$$

$$\Rightarrow m = -\frac{1}{5}$$

Complementary Function: $x = Ate^{-\frac{1}{5}t} + Be^{-\frac{1}{5}t}$

Try this Particular Integral: $x = at^2 + bt + c, \quad \dot{x} = 2at + b, \quad \ddot{x} = 2a$

$$25(2a) + 10(2at + b) + (at^2 + bt + c) = 6 - t^2$$

$$\Rightarrow (a)t^2 + (20a + b)t + (50a + 10b + c) = -t^2 + 6$$

Compare coefficients: $a = -1$

$$20(-1) + b = 0 \Rightarrow b = 20$$

$$50(-1) + 10(20) + c = 6 \Rightarrow c = -144$$

Particular Integral: $x = -t^2 + 20t - 144$

General Solution: $x = Ate^{-\frac{1}{5}t} + Be^{-\frac{1}{5}t} - t^2 + 20t - 144.$

13)

$$\frac{d^2y}{dx^2} + 25y = 50x^2 + 25x$$

Homogeneous Equation: $\frac{d^2y}{dx^2} + 25y = 0$

Auxiliary Equation: $m^2 + 25 = 0$

$$\Rightarrow m^2 = -25$$

$$\Rightarrow m = \pm 5i$$

Complementary Function: $y = A\cos 5x + B\sin 5x$

Try this Particular Integral: $y = ax^2 + bx + c, \quad y' = 2ax + b, \quad y'' = 2a$

$$\Rightarrow 2a + 25(ax^2 + bx + c) = 50x^2 + 25x$$

$$\Rightarrow (25a)x^2 + (25b)x + (2a + 25c) = 50x^2 + 25x$$

Compare coefficients: $25a = 50 \Rightarrow a = 2$

$$25b = 25 \Rightarrow b = 1$$

$$2(2) + 25c = 0 \Rightarrow c = -\frac{4}{25}$$

Particular Integral: $y = 2x^2 + x - \frac{4}{25}$

General Solution: $y = A\cos 5x + B\sin 5x + 2x^2 + x - \frac{4}{25}$

Substitute $y = 0$ and $x = 0$: $\quad 0 = A - \frac{4}{25} \Rightarrow A = \frac{4}{25}$

$$\therefore y = \frac{4}{25}\cos 5x + B\sin 5x + 2x^2 + x - \frac{4}{25}$$

$$\frac{dy}{dx} = -\frac{4}{5}\sin 5x + 5B\cos 5x + 4x + 1$$

Substitute $\frac{dy}{dx} = 1$ and $x = 0$: $\quad 1 = 5B + 1 \Rightarrow B = 0$

Particular Solution: $y = \frac{4}{25}\cos 5x + 2x^2 + x - \frac{4}{25}.$

14)

$$4\ddot{x} - 4\dot{x} + x = 4t + 5$$

$$
\begin{aligned}
\text{Homogeneous Equation: } & 4\ddot{x} - 4\dot{x} + x = 0 \\
\text{Auxiliary Equation: } & 4m^2 - 4m + 1 = 0 \\
& \Rightarrow (2m-1)^2 = 0 \\
& \Rightarrow m = \frac{1}{2} \\
\text{Complementary Function: } & x = Ate^{\frac{1}{2}t} + Be^{\frac{1}{2}t} \\
\text{Try this Particular Integral: } & x = at + b, \quad \dot{x} = a, \quad \ddot{x} = 0 \\
& \Rightarrow 0 - 4a + (at + b) = 4t + 5 \\
& \Rightarrow (a)t + (-4a + b) = 4t + 5 \\
\text{Compare coefficients: } & a = 4 \\
& -4(4) + b = 5 \Rightarrow b = 21 \\
\text{Particular Integral: } & x = 4t + 21 \\
\text{General Solution: } & x = Ate^{\frac{1}{2}t} + Be^{\frac{1}{2}t} + 4t + 21 \\
\text{Substitute } x = 1 \text{ and } t = 0: \quad & 1 = B + 21 \Rightarrow B = -20 \\
& \therefore x = Ate^{\frac{1}{2}t} - 20e^{\frac{1}{2}t} + 4t + 21 \\
& \dot{x} = \frac{1}{2}Ate^{\frac{1}{2}t} + Ae^{\frac{1}{2}t} - 10e^{\frac{1}{2}t} + 4 \\
\text{Substitute } \dot{x} = 0 \text{ and } t = 0: \quad & 0 = A - 10 + 4 \Rightarrow A = 6 \\
\text{Particular Solution: } & x = 6te^{\frac{1}{2}t} - 20e^{\frac{1}{2}t} + 4t + 21.
\end{aligned}
$$

15)

$$
\begin{aligned}
& y = ke^x, \quad \frac{dy}{dx} = ke^x, \quad \frac{d^2y}{dx^2} = ke^x \\
& \frac{d^2y}{dx^2} - 9\frac{dy}{dx} + 14y = 12e^x \\
\Rightarrow\ & ke^x - 9(ke^x) + 14(ke^x) = 12e^x \\
& \Rightarrow (1 - 9 + 14)ke^x = 12e^x \\
& \Rightarrow 6k = 12 \\
& \Rightarrow k = 2 \\
\text{Particular Integral: } & y = 2e^x \\
& \therefore k = 2.
\end{aligned}
$$

16)

$$
\begin{aligned}
& y = ke^{2x}, \quad \frac{dy}{dx} = 2ke^{2x}, \quad \frac{d^2y}{dx^2} = 4ke^{2x} \\
& \frac{d^2y}{dx^2} - \frac{dy}{dx} - 12y = 8e^{2x} \\
\Rightarrow\ & 4ke^{2x} - (2ke^{2x}) - 12(ke^{2x}) = 8e^{2x} \\
& \Rightarrow (4 - 2 - 12)ke^{2x} = 8e^{2x} \\
& \Rightarrow -10k = 8 \\
& \Rightarrow k = -\frac{4}{5}
\end{aligned}
$$

$$\text{Particular Integral: } y = -\frac{4}{5}e^{2x}$$
$$\therefore k = -\frac{4}{5}.$$

17 a)

$$y = ke^{4x}, \quad y' = 4ke^{4x}, \quad y'' = 16ke^{4x}$$
$$\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 5e^{4x}$$
$$\Rightarrow 16ke^{4x} - 3(4ke^{4x}) + 2(ke^{4x}) = 5e^{4x}$$
$$\Rightarrow (16 - 12 + 2)ke^{4x} = 5e^{4x}$$
$$\Rightarrow 6k = 5 \Rightarrow k = \frac{5}{6}$$
$$\text{Particular Integral: } y = \frac{5}{6}e^{4x}.$$

17 b)

$$y = ke^{-2x}, \quad y' = -2ke^{-2x}, \quad y'' = 4ke^{-2x}$$
$$y'' + y' - 6y = 3e^{-2x}$$
$$\Rightarrow 4ke^{-2x} + (-2ke^{-2x}) - 6(ke^{-2x}) = 3e^{-2x}$$
$$\Rightarrow (4 - 2 - 6)ke^{-2x} = 3e^{-2x}$$
$$\Rightarrow -4k = 3 \Rightarrow k = -\frac{3}{4}$$
$$\text{Particular Integral: } y = -\frac{3}{4}e^{-2x}.$$

17 c)

$$y = ke^{x}, \quad y' = ke^{x}, \quad y'' = ke^{x}$$
$$2\frac{d^2y}{dx^2} - 5\frac{dy}{dx} - 3y = 4e^{x}$$
$$\Rightarrow 2(ke^{x}) - 5(ke^{x}) - 3(ke^{x}) = 4e^{x}$$
$$\Rightarrow (2 - 5 - 3)ke^{x} = 4e^{x}$$
$$\Rightarrow -6k = 4 \Rightarrow k = -\frac{2}{3}$$
$$\text{Particular Integral: } y = -\frac{2}{3}e^{x}.$$

18 a)

$$y'' - 3y' + 2y = 5e^{4x}$$
$$\text{Homogeneous Equation: } y'' - 3y' + 2y = 0$$

$$\begin{aligned}
\text{Auxiliary Equation: } & m^2 - 3m + 2 = 0 \\
\Rightarrow & (m-1)(m-2) = 0 \\
\Rightarrow & m = 1 \text{ or } m = 2 \\
\text{Complementary Function: } & y = Ae^x + Be^{2x} \\
\text{Particular Integral: } & y = \frac{5}{6}e^{4x} \\
\text{General Solution: } & y = Ae^x + Be^{2x} + \frac{5}{6}e^{4x}.
\end{aligned}$$

18 b)

$$\begin{aligned}
& y'' + y' - 6y = 3e^{-2x} \\
\text{Homogeneous Equation: } & y'' + y' - 6y = 0 \\
\text{Auxiliary Equation: } & m^2 + m - 6 = 0 \\
\Rightarrow & (m+3)(m-2) = 0 \\
\Rightarrow & m = -3 \text{ or } m = 2 \\
\text{Complementary Function: } & y = Ae^{-3x} + Be^{2x} \\
\text{Particular Integral: } & y = -\frac{3}{4}e^{-2x} \\
\text{General Solution: } & y = Ae^{-3x} + Be^{2x} - \frac{3}{4}e^{-2x}.
\end{aligned}$$

18 c)

$$\begin{aligned}
& 2y'' - 5y' - 3y = 4e^x \\
\text{Homogeneous Equation: } & 2y'' - 5y' - 3y = 0 \\
\text{Auxiliary Equation: } & 2m^2 - 5m - 3 = 0 \\
\Rightarrow & (2m+1)(m-3) = 0 \\
\Rightarrow & m = -\frac{1}{2} \text{ or } m = 3 \\
\text{Complementary Function: } & y = Ae^{-\frac{1}{2}x} + Be^{3x} \\
\text{Particular Integral: } & y = -\frac{2}{3}e^x \\
\text{General Solution: } & y = Ae^{-\frac{1}{2}x} + Be^{3x} - \frac{2}{3}e^x.
\end{aligned}$$

19 a)

$$\begin{aligned}
\text{Homogeneous Equation: } & y'' - 8y' + 15y = 0 \\
\text{Auxiliary Equation: } & m^2 - 8m + 15 = 0 \\
\Rightarrow & (m-3)(m-5) = 0 \\
\Rightarrow & m = 3 \text{ or } m = 5 \\
\text{Complementary Function: } & y = Ae^{3x} + Be^{5x}.
\end{aligned}$$

19 b)

$$y = ke^{3x}, \quad y' = 3ke^{3x}, \quad y'' = 9ke^{3x}$$
$$y'' - 8y' + 15y = 6e^{3x}$$
$$\Rightarrow 9ke^{3x} - 8(3ke^{3x}) + 15(ke^{3x}) = 6e^{3x}$$
$$\Rightarrow (9 - 24 + 15)ke^{3x} = 6e^{3x}$$
$$\Rightarrow 0 = 6$$

There is no value of k for which this is true.

19 c)

$$y = kxe^{3x}, \quad y' = 3kxe^{3x} + ke^{3x}$$
$$y'' = 9kxe^{3x} + 3ke^{3x} + 3ke^{3x}$$
$$= 9kxe^{3x} + 6ke^{3x}$$
$$y'' - 8y' + 15y = 6e^{3x}$$
$$\Rightarrow (9kxe^{3x} + 6ke^{3x}) - 8(3kxe^{3x} + ke^{3x}) + 15(kxe^{3x}) = 6e^{3x}$$
$$\Rightarrow (9 - 24 + 15)kxe^{3x} + (6 - 8)ke^{3x} = 6e^{3x}$$
$$\Rightarrow 0 - 2ke^{3x} = 6e^{3x}$$
$$\Rightarrow -2k = 6 \Rightarrow k = -3$$

Particular Integral: $y = -3xe^{3x}$

$$\therefore k = -3.$$

19 d)

General Solution: $y = Ae^{3x} + Be^{5x} - 3xe^{3x}$.

20 a)

$$2\ddot{x} - 5\dot{x} - 3x = 6e^{3t}$$

Homogeneous Equation: $2\ddot{x} - 5\dot{x} - 3x = 0$

Auxiliary Equation: $2m^2 - 5m - 3 = 0$

$$\Rightarrow (2m + 1)(m - 3) = 0$$
$$\Rightarrow m = -\frac{1}{2} \text{ or } m = 3$$

Complementary Function: $x = Ae^{-\frac{1}{2}t} + Be^{3t}$

Try this Particular Integral: $x = kte^{3t}$

$$\dot{x} = 3kte^{3t} + ke^{3t}$$
$$\ddot{x} = 9kte^{3t} + 3ke^{3t} + 3ke^{3t} = 9kte^{3t} + 6ke^{3t}$$
$$2(9kte^{3t} + 6ke^{3t}) - 5(3kte^{3t} + ke^{3t}) - 3(kte^{3t}) = 6e^{3t}$$

$$\Rightarrow (18-15-3)kte^{3t} + (12-5)ke^{3t} = 6e^{3t}$$
$$\Rightarrow 7ke^{3t} = 6e^{3t} \Rightarrow k = \frac{6}{7}$$
$$\text{Particular Integral: } x = \frac{6}{7}te^{3t}$$
$$\text{General Solution: } x = Ae^{-\frac{1}{2}t} + Be^{3t} + \frac{6}{7}te^{3t}.$$

20 b)

$$4y'' - 4y' + y = 5e^x$$
$$\text{Homogeneous Equation: } 4y'' - 4y' + y = 0$$
$$\text{Auxiliary Equation: } 4m^2 - 4m + 1 = 0$$
$$\Rightarrow (2m-1)^2 = 0 \Rightarrow m = \frac{1}{2}$$
$$\text{Complementary Function: } y = Axe^{\frac{1}{2}x} + Be^{\frac{1}{2}x}$$
$$\text{Try this Particular Integral: } y = ke^x, \quad y' = ke^x, \quad y'' = ke^x$$
$$4(ke^x) - 4(ke^x) + (ke^x) = 5e^x \Rightarrow k = 5$$
$$\text{Particular Integral: } y = 5e^x$$
$$\text{General Solution: } y = Axe^{\frac{1}{2}x} + Be^{\frac{1}{2}x} + 5e^x.$$

20 c)

$$4\ddot{x} + 81x = 8e^{\frac{1}{2}t}$$
$$\text{Homogeneous Equation: } 4\ddot{x} + 81x = 0$$
$$\text{Auxiliary Equation: } 4m^2 + 81 = 0$$
$$\Rightarrow m^2 = -\frac{81}{4}$$
$$\Rightarrow m = \pm\frac{9}{2}i$$
$$\text{Complementary Function: } x = A\cos\frac{9}{2}t + B\sin\frac{9}{2}t$$
$$\text{Try this Particular Integral: } x = ke^{\frac{1}{2}t}, \quad \dot{x} = \frac{1}{2}ke^{\frac{1}{2}t}, \quad \ddot{x} = \frac{1}{4}ke^{\frac{1}{2}t}$$
$$4(\frac{1}{4}ke^{\frac{1}{2}t}) + 81(ke^{\frac{1}{2}t}) = 8e^{\frac{1}{2}t}$$
$$\Rightarrow 82k = 8 \Rightarrow k = \frac{8}{82} = \frac{4}{41}$$
$$\text{Particular Integral: } x = \frac{4}{41}e^{\frac{1}{2}t}$$
$$\text{General Solution: } x = A\cos\frac{9}{2}t + B\sin\frac{9}{2}t + \frac{4}{41}e^{\frac{1}{2}t}.$$

20 d)

$$9y'' - 24y' + 52y = 17e^{-x}$$

$$\text{Homogeneous Equation: } 9y'' - 24y' + 52y = 0$$
$$\text{Auxiliary Equation: } 9m^2 - 24m + 52 = 0$$
$$\Rightarrow 9(m^2 - \frac{8}{3}m) + 52 = 0$$
$$\Rightarrow 9((m - \frac{4}{3})^2 - \frac{16}{9}) + 52 = 0$$
$$\Rightarrow 9(m - \frac{4}{3})^2 - 16 + 52 = 0$$
$$\Rightarrow (m - \frac{4}{3})^2 = -4$$
$$\Rightarrow m = \frac{4}{3} \pm 2i$$
$$\text{Complementary Function: } y = e^{\frac{4}{3}x}(A\cos 2x + B\sin 2x)$$
$$\text{Try this Particular Integral: } y = ke^{-x}, \quad y' = -ke^{-x}, \quad y'' = ke^{-x}$$
$$9(ke^{-x}) - 24(-ke^{-x}) + 52(ke^{-x}) = 17e^{-x}$$
$$\Rightarrow 85k = 17 \Rightarrow k = \frac{1}{5}$$
$$\text{Particular Integral: } y = \frac{1}{5}e^{-x}$$
$$\text{General Solution: } y = e^{\frac{4}{3}x}(A\cos 2x + B\sin 2x) + \frac{1}{5}e^{-x}.$$

21)

$$\ddot{x} + 9x = 9e^{-2t}$$
$$\text{Homogeneous Equation: } \ddot{x} + 9x = 0$$
$$\text{Auxiliary Equation: } m^2 + 9 = 0 \Rightarrow m^2 = -9 \Rightarrow m = \pm 3i$$
$$\text{Complementary Function: } x = A\cos 3t + B\sin 3t$$
$$\text{Try this Particular Integral: } x = ke^{-2t}, \quad \dot{x} = -2ke^{-2t}, \quad \ddot{x} = 4ke^{-2t}$$
$$4ke^{-2t} + 9(ke^{-2t}) = 9e^{-2t}$$
$$\Rightarrow (4+9)ke^{-2t} = 9e^{-2t} \Rightarrow 13k = 9 \Rightarrow k = \frac{9}{13}$$
$$\text{Particular Integral: } x = \frac{9}{13}e^{-2t}$$
$$\text{General Solution: } x = A\cos 3t + B\sin 3t + \frac{9}{13}e^{-2t}$$
$$\text{Substitute } x = 0, t = 0: \quad 0 = A + \frac{9}{13} \Rightarrow A = -\frac{9}{13}$$
$$\therefore x = -\frac{9}{13}\cos 3t + B\sin 3t + \frac{9}{13}e^{-2t}$$
$$\dot{x} = \frac{27}{13}\sin 3t + 3B\cos 3t - \frac{18}{13}e^{-2t}$$
$$\text{Substitute } \dot{x} = 9, t = 0: \quad 9 = 3B - \frac{18}{13} \Rightarrow B = \frac{45}{13}$$
$$\text{Particular Solution: } x = -\frac{9}{13}\cos 3t + \frac{45}{13}\sin 3t + \frac{9}{13}e^{-2t}.$$

22)

$$
\begin{aligned}
y'' + y' - 12y &= 60e^{-3x} \\
\text{Homogeneous Equation: } y'' + y' - 12y &= 0 \\
\text{Auxiliary Equation: } m^2 + m - 12 &= 0 \\
\Rightarrow (m+4)(m-3) &= 0 \Rightarrow m = -4 \text{ or } m = 3 \\
\text{Complementary Function: } y &= Ae^{-4x} + Be^{3x} \\
\text{Try this Particular Integral: } y &= ke^{-3x}, \quad y' = -3ke^{-3x}, \quad y'' = 9ke^{-3x} \\
9ke^{-3x} + (-3ke^{-3x}) - 12ke^{-3x} &= 60e^{-3x} \\
\Rightarrow (9 - 3 - 12)ke^{-3x} &= 60e^{-3x} \Rightarrow -6k = 60 \Rightarrow k = -10 \\
\text{Particular Integral: } y &= -10e^{-3x} \\
\text{General Solution: } y &= Ae^{-4x} + Be^{3x} - 10e^{-3x} \\
\text{As } x \to \infty, y \to 0. \text{ This can only be true if } B &= 0, \text{ otherwise } e^{3x} \to \infty \\
y &= Ae^{-4x} - 10e^{-3x}, \quad y' = -4Ae^{-4x} + 30e^{-3x} \\
\text{Substitute } y' = 0, x = 0: \quad 0 &= -4A + 30 \Rightarrow A = \frac{15}{2} \\
\text{Particular Solution: } y &= \frac{15}{2}e^{-4x} - 10e^{-3x}.
\end{aligned}
$$

23 a)

$$
\begin{aligned}
\text{Homogeneous Equation: } y'' - 4y' + 4y &= 0 \\
\text{Auxiliary Equation: } m^2 - 4m + 4 &= 0 \\
\Rightarrow (m-2)^2 &= 0 \Rightarrow m = 2 \\
\text{Complementary Function: } y &= Axe^{2x} + Be^{2x}.
\end{aligned}
$$

23 b)

$$
\begin{aligned}
y &= ke^{2x}, \quad y' = 2ke^{2x}, \quad y'' = 4ke^{2x} \\
y'' - 4y' + 4y &= 5e^{2x} \\
\Rightarrow 4ke^{2x} - 4(2ke^{2x}) + 4(ke^{2x}) &= 5e^{2x} \\
\Rightarrow (4 - 8 + 4)ke^{2x} &= 5e^{2x} \Rightarrow 0 = 5
\end{aligned}
$$

There is no value of k for which this is true.

23 c)

$$
\begin{aligned}
y &= kxe^{2x}, \quad y' = 2kxe^{2x} + ke^{2x} \\
y'' &= 4kxe^{2x} + 2ke^{2x} + 2ke^{2x} = 4kxe^{2x} + 4ke^{2x} \\
y'' - 4y' + 4y &= 5e^{2x}
\end{aligned}
$$

$$\Rightarrow (4kxe^{2x} + 4ke^{2x}) - 4(2kxe^{2x} + ke^{2x}) + 4(kxe^{2x}) = 5e^{2x}$$
$$\Rightarrow (4 - 8 + 4)kxe^{2x} + (4 - 4)ke^{2x} = 5e^{2x}$$
$$\Rightarrow 0 + 0 = 5$$

There is no value of k for which this is true.

23 d)

$$y = kx^2e^{2x}, \quad y' = 2kx^2e^{2x} + 2kxe^{2x}$$
$$y'' = 4kx^2e^{2x} + 4kxe^{2x} + 4kxe^{2x} + 2ke^{2x}$$
$$= 4kx^2e^{2x} + 8kxe^{2x} + 2ke^{2x}$$
$$y'' - 4y' + 4y = 5e^{2x}$$
$$\Rightarrow (4kx^2e^{2x} + 8kxe^{2x} + 2ke^{2x}) - 4(2kx^2e^{2x} + 2kxe^{2x}) + 4(kx^2e^{2x}) = 5e^{2x}$$
$$\Rightarrow (4 - 8 + 4)kx^2e^{2x} + (8 - 8)kxe^{2x} + 2ke^{2x} = 5e^{2x}$$
$$\Rightarrow 2k = 5 \Rightarrow k = \frac{5}{2}$$

Particular Integral: $y = \frac{5}{2}x^2e^{2x}$

$$\therefore k = \frac{5}{2}.$$

23 e)

General Solution: $y = Axe^{2x} + Be^{2x} + \frac{5}{2}x^2e^{2x}$.

24)

$$y = p\cos 3x + q\sin 3x$$
$$y' = -3p\sin 3x + 3q\cos 3x$$
$$y'' = -9p\cos 3x - 9q\sin 3x$$
$$y'' + 10y' + 9y = \sin 3x$$
$$\Rightarrow (-9p\cos 3x - 9q\sin 3x) + 10(-3p\sin 3x + 3q\cos 3x) + 9(p\cos 3x + q\sin 3x) = \sin 3x$$
$$\Rightarrow (-9p + 30q + 9p)\cos 3x + (-9q - 30p + 9q)\sin 3x = \sin 3x$$
$$\Rightarrow 30q\cos 3x - 30p\sin 3x = \sin 3x$$

Compare coefficients: $30q = 0 \Rightarrow q = 0$

$$-30p = 1 \Rightarrow p = -\frac{1}{30}$$

Particular Integral: $y = -\frac{1}{30}\cos 3x.$

25)

$$
\begin{aligned}
y &= p\cos 4x + q\sin 4x \\
y' &= -4p\sin 4x + 4q\cos 4x \\
y'' &= -16p\cos 4x - 16q\sin 4x \\
y'' + 9y' - 36y &= 5\cos 4x \\
\Rightarrow (-16p + 36q - 36p)\cos 4x + (-16q - 36p - 36q)\sin 4x &= 5\cos 4x \\
\Rightarrow (-52p + 36q)\cos 4x + (-36p - 52q)\sin 4x &= 5\cos 4x \\
-52p + 36q &= 5 \quad (1) \\
-36p - 52q &= 0 \quad (2) \\
\text{Solve (1) and (2): } p &= -\frac{13}{200}, \quad q = \frac{9}{200} \\
\text{Particular Integral: } y &= -\frac{13}{200}\cos 4x + \frac{9}{200}\sin 4x.
\end{aligned}
$$

26 a)

$$
\begin{aligned}
y &= p\cos 6x + q\sin 6x \\
y' &= -6p\sin 6x + 6q\cos 6x \\
y'' &= -36p\cos 6x - 36q\sin 6x \\
25y'' - 9y &= 9\cos 6x \\
\Rightarrow 25(-36p\cos 6x - 36q\sin 6x) - 9(p\cos 6x + q\sin 6x) & \\
&= 9\cos 6x \\
\Rightarrow (-900p - 9p)\cos 6x + (-900q - 9q)\sin 6x &= 9\cos 6x \\
\Rightarrow -909p\cos 6x - 909q\sin 6x &= 9\cos 6x \\
\text{Compare coefficients: } -909p &= 9 \Rightarrow p = -\frac{1}{101} \\
-909q &= 0 \Rightarrow q = 0 \\
\text{Particular Integral: } y &= -\frac{1}{101}\cos 6x.
\end{aligned}
$$

26 b)

$$
\begin{aligned}
y &= p\cos 8x + q\sin 8x \\
y' &= -8p\sin 8x + 8q\cos 8x \\
y'' &= -64p\cos 8x - 64q\sin 8x \\
2y'' - 11y' + 5y &= 89\sin 8x \\
\Rightarrow 2(-64p\cos 8x - 64q\sin 8x) - 11(-8p\sin 8x + 8q\cos 8x) & \\
+ 5(p\cos 8x + q\sin 8x) &= 89\sin 8x \\
\Rightarrow (-128p - 88q + 5p)\cos 8x + (-128q + 88p + 5q)\sin 8x &= 89\sin 8x \\
\Rightarrow (-123p - 88q)\cos 8x + (88p - 123q)\sin 8x &= 89\sin 8x \\
-123p - 88q = 0 \quad (1), \qquad 88p - 123q &= 89 \quad (2) \\
\text{Solve (1) and (2): } p &= \frac{88}{257}, \quad q = -\frac{123}{257}
\end{aligned}
$$

$$\text{Particular Integral: } y = \frac{88}{257}\cos 8x - \frac{123}{257}\sin 8x.$$

26 c)

$$\begin{aligned}
y &= p\cos 2x + q\sin 2x \\
y' &= -2p\sin 2x + 2q\cos 2x \\
y'' &= -4p\cos 2x - 4q\sin 2x \\
3y'' + 11y' + 6y &= 20\sin 2x - 4\cos 2x \\
\Rightarrow 3(-4p\cos 2x - 4q\sin 2x) & \\
+ 11(-2p\sin 2x + 2q\cos 2x) & \\
+ 6(p\cos 2x + q\sin 2x) &= 20\sin 2x - 4\cos 2x \\
\Rightarrow (-12p + 22q + 6p)\cos 2x & \\
+ (-12q - 22p + 6q)\sin 2x &= 20\sin 2x - 4\cos 2x \\
\Rightarrow (-6p + 22q)\cos 2x & \\
+ (-22p - 6q)\sin 2x &= 20\sin 2x - 4\cos 2x \\
-6p + 22q &= -4 \quad (1), \qquad -22p - 6q = 20 \quad (2) \\
\text{Solve (1) and (2): } p &= -\frac{4}{5}, \quad q = -\frac{2}{5} \\
\text{Particular Integral: } y &= -\frac{4}{5}\cos 2x - \frac{2}{5}\sin 2x.
\end{aligned}$$

27 a)

$$\begin{aligned}
25y'' - 9y &= 9\cos 6x \\
\text{Homogeneous Equation: } 25y'' - 9y &= 0 \\
\text{Auxiliary Equation: } 25m^2 - 9 &= 0 \\
\Rightarrow (5m-3)(5m+3) = 0 &\Rightarrow m = \frac{3}{5} \text{ or } m = -\frac{3}{5} \\
\text{Complementary Function: } y &= Ae^{\frac{3}{5}x} + Be^{-\frac{3}{5}x} \\
\text{Particular Integral: } y &= -\frac{1}{101}\cos 6x \\
\text{General Solution: } y &= Ae^{\frac{3}{5}x} + Be^{-\frac{3}{5}x} - \frac{1}{101}\cos 6x.
\end{aligned}$$

27 b)

$$\begin{aligned}
2y'' - 11y' + 5y &= 89\sin 8x \\
\text{Homogeneous Equation: } 2y'' - 11y' + 5y &= 0 \\
\text{Auxiliary Equation: } 2m^2 - 11m + 5 &= 0
\end{aligned}$$

$$\Rightarrow (2m-1)(m-5)=0 \Rightarrow m=\frac{1}{2} \text{ or } m=5$$

$$\text{Complementary Function: } y=Ae^{\frac{1}{2}x}+Be^{5x}$$

$$\text{Particular Integral: } y=\frac{88}{257}\cos 8x-\frac{123}{257}\sin 8x$$

$$\text{General Solution: } y=Ae^{\frac{1}{2}x}+Be^{5x}+\frac{88}{257}\cos 8x-\frac{123}{257}\sin 8x.$$

27 c)

$$3y''+11y'+6y=20\sin 2x-4\cos 2x$$

$$\text{Homogeneous Equation: } 3y''+11y'+6y=0$$

$$\text{Auxiliary Equation: } 3m^2+11m+6=0$$

$$\Rightarrow (3m+2)(m+3)=0 \Rightarrow m=-\frac{2}{3} \text{ or } m=-3$$

$$\text{Complementary Function: } y=Ae^{-\frac{2}{3}x}+Be^{-3x}$$

$$\text{Particular Integral: } y=-\frac{4}{5}\cos 2x-\frac{2}{5}\sin 2x$$

$$\text{General Solution: } y=Ae^{-\frac{2}{3}x}+Be^{-3x}-\frac{4}{5}\cos 2x-\frac{2}{5}\sin 2x.$$

28 a)

$$\text{Homogeneous Equation: } y''+4y=0$$

$$\text{Auxiliary Equation: } m^2+4=0 \Rightarrow m^2=-4 \Rightarrow m=\pm 2i$$

$$\text{Complementary Function: } y=A\cos 2x+B\sin 2x.$$

28 b)

$$y=p\cos 2x+q\sin 2x$$

$$y'=-2p\sin 2x+2q\cos 2x$$

$$y''=-4p\cos 2x-4q\sin 2x$$

$$y''+4y=\sin 2x$$

$$\Rightarrow (-4p\cos 2x-4q\sin 2x)+4(p\cos 2x+q\sin 2x)=\sin 2x$$

$$\Rightarrow (-4p+4p)\cos 2x+(-4q+4q)\sin 2x=\sin 2x$$

$$\Rightarrow 0+0=1$$

There is no value of k for which this is true.

28 c)

$$y=px\cos 2x+qx\sin 2x$$

$$y'=-2px\sin 2x+p\cos 2x+2qx\cos 2x+q\sin 2x$$

$$
\begin{aligned}
y'' &= -4px\cos 2x - 2p\sin 2x - 2p\sin 2x \\
&\quad - 4qx\sin 2x + 2q\cos 2x + 2q\cos 2x \\
&= -4px\cos 2x - 4p\sin 2x - 4qx\sin 2x + 4q\cos 2x \\
y'' + 4y &= \sin 2x \\
\Rightarrow (-4px\cos 2x - 4p\sin 2x & \\
&\quad - 4qx\sin 2x + 4q\cos 2x) \\
&\quad + 4(px\cos 2x + qx\sin 2x) = \sin 2x \\
\Rightarrow (-4p + 4p)x\cos 2x + 4q\cos 2x & \\
&\quad + (-4q + 4q)x\sin 2x - 4p\sin 2x = \sin 2x \\
\Rightarrow 4q\cos 2x - 4p\sin 2x &= \sin 2x \\
4q = 0 \Rightarrow q &= 0 \\
-4p = 1 \Rightarrow p &= -\frac{1}{4} \\
\text{Particular Integral: } y &= -\frac{1}{4}x\cos 2x \\
\therefore p = -\frac{1}{4} &\text{ and } q = 0.
\end{aligned}
$$

28 d)

$$\text{General Solution: } y = A\cos 2x + B\sin 2x - \frac{1}{4}x\cos 2x.$$

29 a)

$$y = k.$$

29 b)

$$y = ke^{3x}.$$

29 c)

$$y = ax + b.$$

29 d)

$$y = p\sin x + q\cos x.$$

29 e)

$$y = ax^2 + bx + c.$$

29 f)

$$y = kxe^{-x}.$$

29 g)

$$y = p\cos 4x + q\sin 4x.$$

29 h)

$$y = ax + b.$$

29 i)

$$y = p\cos 7x + q\sin 7x.$$

29 j)

$$y = kxe^{-2x}.$$

30 a)

$$y = ke^{-x}.$$

30 b)

$$y = ax + b.$$

30 c)

$$y = p\cos 2x + q\sin 2x.$$

30 d)

$$y = ax^2 + bx + c.$$

30 e)

$$y = k.$$

30 f)

$$y = px\cos 6x + qx\sin 6x.$$

30 g)

$$y = ke^{6x}.$$

30 h)

$$y = px\cos 6x + qx\sin 6x.$$

30 i)

$$y = p\cos 9x + q\sin 9x.$$

30 j)

$$y = px\cos 6x + qx\sin 6x.$$

31 a)

$$\begin{aligned}
\text{Homogeneous Equation: } & y'' + 9y' - 52y = 0 \\
\text{Auxiliary Equation: } & m^2 + 9m - 52 = 0 \\
& \Rightarrow (m+13)(m-4) = 0 \Rightarrow m = -13 \text{ or } m = 4 \\
\text{Complementary Function: } & y = Ae^{-13x} + Be^{4x} \\
\text{Try this Particular Integral: } & y = ax + b, \quad y' = a, \quad y'' = 0 \\
& 0 + 9a - 52(ax + b) = 12 - 13x \\
& \Rightarrow (-52a)x + (9a - 52b) = -13x + 12
\end{aligned}$$

$$-52a = -13 \Rightarrow a = \frac{1}{4}$$
$$9(\frac{1}{4}) - 52b = 12 \Rightarrow b = -\frac{3}{16}$$
$$\text{Particular Integral: } y = \frac{1}{4}x - \frac{3}{16}$$
$$\text{General Solution: } y = Ae^{-13x} + Be^{4x} + \frac{1}{4}x - \frac{3}{16}.$$

31 b)

$$\text{Try this Particular Integral: } y = ke^{5x}, \quad y' = 5ke^{5x}, \quad y'' = 25ke^{5x}$$
$$25ke^{5x} + 9(5ke^{5x}) - 52(ke^{5x}) = 3e^{5x}$$
$$\Rightarrow (25 + 45 - 52)k = 3 \Rightarrow 18k = 3 \Rightarrow k = \frac{1}{6}$$
$$\text{Particular Integral: } y = \frac{1}{6}e^{5x}$$
$$\text{General Solution: } y = Ae^{-13x} + Be^{4x} + \frac{1}{6}e^{5x}.$$

31 c)

$$\text{Try this Particular Integral: } y = kxe^{4x}, \quad y' = 4kxe^{4x} + ke^{4x}$$
$$y'' = 16kxe^{4x} + 4ke^{4x} + 4ke^{4x} = 16kxe^{4x} + 8ke^{4x}$$
$$(16kxe^{4x} + 8ke^{4x}) + 9(4kxe^{4x} + ke^{4x}) - 52(kxe^{4x}) = 17e^{4x}$$
$$\Rightarrow (16 + 36 - 52)kxe^{4x} + (8 + 9)ke^{4x} = 17e^{4x}$$
$$\Rightarrow 17k = 17 \Rightarrow k = 1$$
$$\text{Particular Integral: } y = xe^{4x}$$
$$\text{General Solution: } y = Ae^{-13x} + Be^{4x} + xe^{4x}.$$

31 d)

$$\text{Try this Particular Integral: } y = p\cos 3x + q\sin 3x$$
$$y' = -3p\sin 3x + 3q\cos 3x$$
$$y'' = -9p\cos 3x - 9q\sin 3x$$
$$y'' + 9y' - 52y = 89\cos 3x + 89\sin 3x$$
$$\Rightarrow (-61p + 27q)\cos 3x + (-27p - 61q)\sin 3x = 89\cos 3x + 89\sin 3x$$
$$-61p + 27q = 89 \quad (1), \qquad -27p - 61q = 89 \quad (2)$$
$$\text{Solve (1) and (2): } p = -\frac{44}{25}, \quad q = -\frac{17}{25}$$
$$\text{Particular Integral: } y = -\frac{44}{25}\cos 3x - \frac{17}{25}\sin 3x$$
$$\text{General Solution: } y = Ae^{-13x} + Be^{4x} - \frac{44}{25}\cos 3x - \frac{17}{25}\sin 3x.$$

32 a)

$$\begin{aligned}
\text{Homogeneous Equation: } & 4y'' - 20y' + 25y = 0 \\
\text{Auxiliary Equation: } & 4m^2 - 20m + 25 = 0 \Rightarrow (2m-5)^2 = 0 \Rightarrow m = \frac{5}{2} \\
\text{Complementary Function: } & y = Axe^{\frac{5}{2}x} + Be^{\frac{5}{2}x} \\
\text{Try this Particular Integral: } & y = p\cos x + q\sin x \\
& y' = -p\sin x + q\cos x \\
& y'' = -p\cos x - q\sin x \\
& 4y'' - 20y' + 25y = 841\cos x \\
\Rightarrow & (21p - 20q)\cos x + (20p + 21q)\sin x = 841\cos x \\
& 21p - 20q = 841 \quad (1), \quad 20p + 21q = 0 \quad (2) \\
\text{Solve (1) and (2): } & p = 21, \quad q = -20 \\
\text{Particular Integral: } & y = 21\cos x - 20\sin x \\
\text{General Solution: } & y = Axe^{\frac{5}{2}x} + Be^{\frac{5}{2}x} \\
& \quad + 21\cos x - 20\sin x.
\end{aligned}$$

32 b)

$$\begin{aligned}
\text{Try this Particular Integral: } & y = ke^x, \quad y' = ke^x, \quad y'' = ke^x \\
& 4ke^x - 20ke^x + 25ke^x = 18e^x \Rightarrow 9k = 18 \Rightarrow k = 2 \\
\text{Particular Integral: } & y = 2e^x \\
\text{General Solution: } & y = Axe^{\frac{5}{2}x} + Be^{\frac{5}{2}x} + 2e^x.
\end{aligned}$$

32 c)

$$\begin{aligned}
\text{Try this Particular Integral: } & y = ax + b, \quad y' = a, \quad y'' = 0 \\
& -20a + 25(ax + b) = 25 - 125x \\
\Rightarrow & (25a)x + (-20a + 25b) = -125x + 25 \\
& 25a = -125 \Rightarrow a = -5 \\
& -20(-5) + 25b = 25 \Rightarrow b = -3 \\
\text{Particular Integral: } & y = -5x - 3 \\
\text{General Solution: } & y = Axe^{\frac{5}{2}x} + Be^{\frac{5}{2}x} - 5x - 3.
\end{aligned}$$

32 d)

$$\begin{aligned}
\text{Try this Particular Integral: } & y = kx^2e^{\frac{5}{2}x} \\
& y' = \frac{5}{2}kx^2e^{\frac{5}{2}x} + 2kxe^{\frac{5}{2}x} \\
& y'' = \frac{25}{4}kx^2e^{\frac{5}{2}x} + 5kxe^{\frac{5}{2}x} + 5kxe^{\frac{5}{2}x} + 2ke^{\frac{5}{2}x}
\end{aligned}$$

$$= \frac{25}{4}kx^2e^{\frac{5}{2}x} + 10kxe^{\frac{5}{2}x} + 2ke^{\frac{5}{2}x}$$

$$4y'' - 20y' + 25y = e^{\frac{5}{2}x}$$

$$\Rightarrow (25 - 50 + 25)kx^2e^{\frac{5}{2}x} + (40 - 40)kxe^{\frac{5}{2}x} + 8ke^{\frac{5}{2}x} = e^{\frac{5}{2}x}$$

$$\Rightarrow 8k = 1 \Rightarrow k = \frac{1}{8}$$

Particular Integral: $y = \frac{1}{8}x^2e^{\frac{5}{2}x}$

General Solution: $y = Axe^{\frac{5}{2}x} + Be^{\frac{5}{2}x} + \frac{1}{8}x^2e^{\frac{5}{2}x}$.

33 a)

Homogeneous Equation: $\ddot{x} + 2\dot{x} + 2x = 0$

Auxiliary Equation: $m^2 + 2m + 2 = 0$

$$\Rightarrow (m+1)^2 - 1 + 2 = 0 \Rightarrow (m+1)^2 = -1 \Rightarrow m = -1 \pm i$$

Complementary Function: $x = e^{-t}(A\cos t + B\sin t)$

Try this Particular Integral: $x = ke^{-t}, \quad \dot{x} = -ke^{-t}, \quad \ddot{x} = ke^{-t}$

$$ke^{-t} + 2(-ke^{-t}) + 2ke^{-t} = 2e^{-t} \Rightarrow k = 2$$

Particular Integral: $x = 2e^{-t}$

General Solution: $x = e^{-t}(A\cos t + B\sin t) + 2e^{-t}$

$$\Rightarrow x = e^{-t}(A\cos t + B\sin t + 2)$$

Substitute $x = 0, t = 0$: $\quad 0 = A + 2 \Rightarrow A = -2$

$$x = e^{-t}(-2\cos t + B\sin t + 2)$$

$$\dot{x} = e^{-t}(2\sin t + B\cos t) - e^{-t}(-2\cos t + B\sin t + 2)$$

Substitute $\dot{x} = 0, t = 0$: $\quad 0 = B - (-2 + 2) \Rightarrow B = 0$

Particular Solution: $x = e^{-t}(-2\cos t + 2) = -2e^{-t}(\cos t - 1)$.

33 b)

Try this Particular Integral: $x = p\cos t + q\sin t$

$$\dot{x} = -p\sin t + q\cos t, \quad \ddot{x} = -p\cos t - q\sin t$$

$$\ddot{x} + 2\dot{x} + 2x = 5\sin t$$

$$\Rightarrow (p + 2q)\cos t + (-2p + q)\sin t = 5\sin t$$

$$p + 2q = 0 \quad (1), \quad -2p + q = 5 \quad (2)$$

Solve (1) and (2): $p = -2, \quad q = 1$

Particular Integral: $x = -2\cos t + \sin t$

General Solution: $x = e^{-t}(A\cos t + B\sin t) - 2\cos t + \sin t$

Substitute $x = 0, t = 0$: $\quad 0 = A - 2 \Rightarrow A = 2$

$$x = e^{-t}(2\cos t + B\sin t) - 2\cos t + \sin t$$

$$\dot{x} = e^{-t}(-2\sin t + B\cos t) - e^{-t}(2\cos t + B\sin t) + 2\sin t + \cos t$$

Substitute $\dot{x} = 0, t = 0$: $\quad 0 = B - 2 + 1 \Rightarrow B = 1$

Particular Solution: $x = e^{-t}(2\cos t + \sin t) - 2\cos t + \sin t$.

33 c)

$$
\begin{aligned}
\text{Try this Particular Integral: } x &= at^2 + bt + c, \quad \dot{x} = 2at + b, \quad \ddot{x} = 2a \\
\ddot{x} + 2\dot{x} + 2x &= 6t^2 \\
2a + 2(2at + b) + 2(at^2 + bt + c) &= 6t^2 \\
\Rightarrow (2a)t^2 + (4a + 2b)t + (2a + 2b + 2c) &= 6t^2 \\
2a = 6 \Rightarrow a &= 3 \\
4(3) + 2b = 0 \Rightarrow b &= -6 \\
2(3) + 2(-6) + 2c = 0 \Rightarrow c &= 3 \\
\text{Particular Integral: } x &= 3t^2 - 6t + 3 \\
\text{General Solution: } x &= e^{-t}(A\cos t + B\sin t) + 3t^2 - 6t + 3 \\
\text{Substitute } x = 0, t = 0: \quad 0 &= A + 3 \Rightarrow A = -3 \\
x &= e^{-t}(-3\cos t + B\sin t) + 3t^2 - 6t + 3 \\
\dot{x} &= e^{-t}(3\sin t + B\cos t) - e^{-t}(-3\cos t + B\sin t) + 6t - 6 \\
\text{Substitute } \dot{x} = 0, t = 0: \quad 0 &= B - (-3) - 6 \Rightarrow B = 3 \\
\text{Particular Solution: } x &= e^{-t}(-3\cos t + 3\sin t) + 3t^2 - 6t + 3.
\end{aligned}
$$

33 d)

$$
\begin{aligned}
\text{Try this Particular Integral: } x &= p\cosh t + q\sinh t \\
\dot{x} &= p\sinh t + q\cosh t, \quad \ddot{x} = p\cosh t + q\sinh t \\
\ddot{x} + 2\dot{x} + 2x &= 4\sinh t \\
\Rightarrow (3p + 2q)\cosh t + (2p + 3q)\sinh t &= 4\sinh t \\
3p + 2q = 0 \quad (1), \quad 2p + 3q &= 4 \quad (2) \\
\text{Solve (1) and (2): } p = -\frac{8}{5}, \quad q &= \frac{12}{5} \\
\text{Particular Integral: } x &= -\frac{8}{5}\cosh t + \frac{12}{5}\sinh t \\
\text{General Solution: } x &= e^{-t}(A\cos t + B\sin t) - \frac{8}{5}\cosh t + \frac{12}{5}\sinh t \\
\text{Substitute } x = 0, t = 0: \quad 0 &= A - \frac{8}{5} \Rightarrow A = \frac{8}{5} \\
x &= e^{-t}(\frac{8}{5}\cos t + B\sin t) - \frac{8}{5}\cosh t + \frac{12}{5}\sinh t \\
\dot{x} &= e^{-t}(-\frac{8}{5}\sin t + B\cos t) - e^{-t}(\frac{8}{5}\cos t + B\sin t) \\
&\quad - \frac{8}{5}\sinh t + \frac{12}{5}\cosh t \\
\text{Substitute } \dot{x} = 0, t = 0: \quad 0 &= B - \frac{8}{5} + \frac{12}{5} \Rightarrow B = -\frac{4}{5} \\
\text{Particular Solution: } x &= e^{-t}(\frac{8}{5}\cos t - \frac{4}{5}\sin t) \\
&\quad - \frac{8}{5}\cosh t + \frac{12}{5}\sinh t.
\end{aligned}
$$

33 e)

$$\begin{aligned}
\text{Try this Particular Integral: } x &= p\cosh 2t + q\sinh 2t \\
\dot{x} &= 2p\sinh 2t + 2q\cosh 2t \\
\ddot{x} &= 4p\cosh 2t + 4q\sinh 2t \\
\ddot{x} + 2\dot{x} + 2x &= 4\sinh t \\
\Rightarrow (4p + 4q + 2p)\cosh 2t + (4q + 4p + 2q)\sinh 2t &= \cosh 2t - \sinh 2t \\
\Rightarrow (6p + 4q)\cosh 2t + (4p + 6q)\sinh 2t &= \cosh 2t - \sinh 2t \\
6p + 4q = 1 \quad (1), \quad 4p + 6q &= -1 \quad (2) \\
\text{Solve (1) and (2): } p = \frac{1}{2}, \quad q &= -\frac{1}{2} \\
\text{Particular Integral: } x &= \frac{1}{2}\cosh 2t - \frac{1}{2}\sinh 2t \\
\text{General Solution: } x &= e^{-t}(A\cos t + B\sin t) + \frac{1}{2}\cosh 2t - \frac{1}{2}\sinh 2t \\
\text{Substitute } x = 0, t = 0: \quad 0 &= A + \frac{1}{2} \Rightarrow A = -\frac{1}{2} \\
x &= e^{-t}(-\frac{1}{2}\cos t + B\sin t) + \frac{1}{2}\cosh 2t - \frac{1}{2}\sinh 2t \\
\dot{x} &= e^{-t}(\frac{1}{2}\sin t + B\cos t) - e^{-t}(-\frac{1}{2}\cos t + B\sin t) \\
&\quad + \sinh 2t - \cosh 2t \\
\text{Substitute } \dot{x} = 0, t = 0: \quad 0 &= B + \frac{1}{2} - 1 \Rightarrow B = \frac{1}{2} \\
\text{Particular Solution: } x &= e^{-t}(-\frac{1}{2}\cos t + \frac{1}{2}\sin t) \\
&\quad + \frac{1}{2}\cosh 2t - \frac{1}{2}\sinh 2t.
\end{aligned}$$

34)

$$\begin{aligned}
y &= pe^{2x} + qe^{4x} \\
y' &= 2pe^{2x} + 4qe^{4x} \\
y'' &= 4pe^{2x} + 16qe^{4x} \\
5y'' + 9y' - 2y &= 4e^{2x} - 6e^{4x} \\
\Rightarrow (20p + 18p - 2p)e^{2x} + (80q + 36q - 2q)e^{4x} &= 4e^{2x} - 6e^{4x} \\
\Rightarrow 36pe^{2x} + 114qe^{4x} &= 4e^{2x} - 6e^{4x} \\
36p = 4 \Rightarrow p = \frac{1}{9}, \quad 114q &= -6 \Rightarrow q = -\frac{1}{19} \\
\text{Particular Integral: } y &= \frac{1}{9}e^{2x} \\
&\quad - \frac{1}{19}e^{4x}.
\end{aligned}$$

35)

$$y = p + qx + r\cos 2x + k\sin 2x$$

$$y' = q - 2r\sin 2x + 2k\cos 2x$$
$$y'' = -4r\cos 2x - 4k\sin 2x$$
$$4y'' + y = 9x + \cos 2x$$
$$\Rightarrow qx + p - 15r\cos 2x - 15k\sin 2x = 9x + \cos 2x$$
$$q = 9, \quad p = 0, \quad -15r = 1 \Rightarrow r = -\frac{1}{15}, \quad -15k = 0 \Rightarrow k = 0$$
$$\text{Particular Integral: } y = 9x - \frac{1}{15}\cos 2x$$
$$\therefore p = 9,\ q = 0,\ r = -\frac{1}{15},\ k = 0.$$

36)

$$\text{Homogeneous Equation: } y'' - y = 0$$
$$\text{Auxiliary Equation: } m^2 - 1 = 0$$
$$\Rightarrow (m-1)(m+1) = 0 \Rightarrow m = 1 \text{ or } m = -1$$
$$\text{Complementary Function: } y = Ae^x + Be^{-x}$$
$$\text{Try this Particular Integral: } y = px + q + m\cos x + n\sin x$$
$$y' = p - m\sin x + n\cos x, \quad y'' = -m\cos x - n\sin x$$
$$y'' - y = x + \sin x$$
$$\Rightarrow (-p)x - q - 2m\cos x - 2n\sin x = x + \sin x$$
$$-p = 1 \Rightarrow p = -1, \quad -q = 0 \Rightarrow q = 0$$
$$-2m = 0 \Rightarrow m = 0, \quad -2n = 1 \Rightarrow n = -\frac{1}{2}$$
$$\text{Particular Integral: } y = -x - \frac{1}{2}\sin x$$
$$\text{General Solution: } y = Ae^x + Be^{-x} - x - \frac{1}{2}\sin x.$$

37 a)

$$y = p + qe^x.$$

37 b)

$$y = pe^{2x} + qe^x.$$

37 c)

$$y = px + q + re^{5x}.$$

37 d)

$$y = pe^{4x} + q\cos x + r\sin x.$$

37 e)

$$y = p\cos 3x + q\sin 3x + rx^2 + sx + t.$$

37 f)

$$y = p + qx + re^{-5x}.$$

37 g)

$$y = p + qx + r\cos\frac{1}{2}x + k\sin\frac{1}{2}x.$$

37 h)

$$y = pxe^{-6x} + q.$$

37 i)

$$y = pe^{6x} + qxe^{7x} + rx + s.$$

37 j)

$$y = pxe^{-6x} + qxe^{7x} + r\cos 3x + k\sin 3x.$$

38 a)

$$y = p\cos 3x + q\sin 3x + r.$$

38 b)

$$y = px + q + re^{-8x}.$$

38 c)

$$y = pe^x + q\cos\frac{2}{3}x + r\sin\frac{2}{3}x.$$

38 d)

$$y = pe^{4x} + qx + r.$$

38 e)

$$y = p\cos 12x + q\sin 12x + r.$$

38 f)

$$y = pe^{3x} + qe^{-2x} + rx^2 + mx + n.$$

38 g)

$$y = p\cos x + q\sin x + m\cos 2x + n\sin 2x.$$

38 h)

$$y = px^2 + qx + r + ke^{10x}.$$

38 i)

$$y = px^2e^{8x} + qx^2 + rx + k.$$

38 j)

$$y = p\cos 9x + q\sin 9x + m\cos 4x + n\sin 4x + ax^2 + bx + c + kx^2e^{8x}.$$

39)

$$
\begin{aligned}
y'' - 5y' + 6y &= 3e^{2x} + 2e^{3x} \\
\text{Homogeneous Equation: } y'' - 5y' + 6y &= 0 \\
\text{Auxiliary Equation: } m^2 - 5m + 6 &= 0 \\
\Rightarrow (m-2)(m-3) = 0 &\Rightarrow m = 2 \text{ or } m = 3 \\
\text{Complementary Function: } y &= Ae^{2x} + Be^{3x} \\
\text{Try this Particular Integral: } y &= pxe^{2x} + qxe^{3x} \\
y' &= 2pxe^{2x} + pe^{2x} + 3qxe^{3x} + qe^{3x} \\
y'' &= 4pxe^{2x} + 4pe^{2x} + 9qxe^{3x} + 6qe^{3x} \\
y'' - 5y' + 6y &= 3e^{2x} + 2e^{3x} \\
\Rightarrow (4p - 10p + 6p)xe^{2x} + (4p - 5p)e^{2x} & \\
+ (9q - 15q + 6q)xe^{3x} + (6q - 5q)e^{3x} &= 3e^{2x} + 2e^{3x} \\
\Rightarrow -pe^{2x} + qe^{3x} &= 3e^{2x} + 2e^{3x} \\
-p = 3 \Rightarrow p = -3, \quad q &= 2 \\
\text{Particular Integral: } y &= -3xe^{2x} + 2xe^{3x} \\
\text{General Solution: } y &= Ae^{2x} + Be^{3x} \\
&\quad - 3xe^{2x} + 2xe^{3x}.
\end{aligned}
$$

40)

$$
\begin{aligned}
25y'' - 40y' + 16y &= 4 - 9e^{-\frac{2}{5}x} - 10e^{\frac{4}{5}x} \\
\text{Homogeneous Equation: } 25y'' - 40y' + 16y &= 0 \\
\text{Auxiliary Equation: } 25m^2 - 40m + 16 = 0 \Rightarrow (5m-4)^2 = 0 &\Rightarrow m = \frac{4}{5} \\
\text{Complementary Function: } y &= Axe^{\frac{4}{5}x} + Be^{\frac{4}{5}x} \\
\text{Try this Particular Integral: } y &= k + pe^{-\frac{2}{5}x} + qx^2e^{\frac{4}{5}x} \\
y' &= -\frac{2}{5}pe^{-\frac{2}{5}x} + \frac{4}{5}qx^2e^{\frac{4}{5}x} + 2qxe^{\frac{4}{5}x} \\
y'' &= \frac{4}{25}pe^{-\frac{2}{5}x} + \frac{16}{25}qx^2e^{\frac{4}{5}x} \\
&\quad + \frac{8}{5}qxe^{\frac{4}{5}x} + \frac{8}{5}qxe^{\frac{4}{5}x} + 2qe^{\frac{4}{5}x} \\
&= \frac{4}{25}pe^{-\frac{2}{5}x} + \frac{16}{25}qx^2e^{\frac{4}{5}x} + \frac{16}{5}qxe^{\frac{4}{5}x} + 2qe^{\frac{4}{5}x} \\
25y'' - 40y' + 16y &= 4 - 9e^{-\frac{2}{5}x} - 10e^{\frac{4}{5}x} \\
\Rightarrow 16k + 36pe^{-\frac{2}{5}x} + (16q - 32q + 16q)x^2e^{\frac{4}{5}x} & \\
+ (80q - 80q)xe^{\frac{4}{5}x} + 50qe^{\frac{4}{5}x} &= 4 - 9e^{-\frac{2}{5}x} - 10e^{\frac{4}{5}x} \\
\Rightarrow 16k + 36pe^{-\frac{2}{5}x} + 50qe^{\frac{4}{5}x} &= 4 - 9e^{-\frac{2}{5}x} - 10e^{\frac{4}{5}x} \\
16k = 4 \Rightarrow k &= \frac{1}{4} \\
36p = -9 \Rightarrow p &= -\frac{1}{4} \\
50q = -10 \Rightarrow q &= -\frac{1}{5}
\end{aligned}
$$

$$\text{Particular Integral: } y = \frac{1}{4} - \frac{1}{4}e^{-\frac{2}{5}x} - \frac{1}{5}x^2 e^{\frac{4}{5}x}$$

$$\text{General Solution: } y = Axe^{\frac{4}{5}x} + Be^{\frac{4}{5}x} + \frac{1}{4} - \frac{1}{4}e^{-\frac{2}{5}x} - \frac{1}{5}x^2 e^{\frac{4}{5}x}.$$

41 a)

$$\text{Homogeneous Equation: } \ddot{x} - 7\dot{x} + 6x = 0$$

$$\text{Auxiliary Equation: } m^2 - 7m + 6 = 0 \Rightarrow (m-1)(m-6) = 0 \Rightarrow m = 1, 6$$

$$\text{Complementary Function: } x = Ae^t + Be^{6t}$$

$$\text{Try this Particular Integral: } x = pt + q + r\cos t + k\sin t$$

$$\dot{x} = p - r\sin t + k\cos t, \quad \ddot{x} = -r\cos t - k\sin t$$

$$\ddot{x} - 7\dot{x} + 6x = 72t + 74\sin t$$

$$\Rightarrow 6pt + (-7p + 6q) + (5r - 7k)\cos t + (7r + 5k)\sin t = 72t + 74\sin t$$

$$6p = 72 \Rightarrow p = 12, \quad -7(12) + 6q = 0 \Rightarrow q = 14$$

$$5r - 7k = 0 \quad (1), \quad 7r + 5k = 74 \quad (2)$$

$$\text{Solve: } r = 7, \quad k = 5$$

$$\text{Particular Integral: } x = 12t + 14 + 7\cos t + 5\sin t$$

$$\text{General Solution: } x = Ae^t + Be^{6t} + 12t + 14 + 7\cos t + 5\sin t$$

$$x(0) = 0 \Rightarrow A + B = -21 \quad (3)$$

$$\dot{x} = Ae^t + 6Be^{6t} + 12 - 7\sin t + 5\cos t$$

$$\dot{x}(0) = 1 \Rightarrow A + 6B = -16 \quad (4)$$

$$\text{Solve (3) and (4): } A = -22, \quad B = 1$$

$$\text{Particular Solution: } x = -22e^t + e^{6t} + 12t + 14 + 7\cos t + 5\sin t.$$

41 b)

$$\text{Try this Particular Integral: } x = pte^t + qe^{2t}$$

$$\dot{x} = pte^t + pe^t + 2qe^{2t}, \quad \ddot{x} = pte^t + 2pe^t + 4qe^{2t}$$

$$\ddot{x} - 7\dot{x} + 6x = 50e^t - 52e^{2t}$$

$$\Rightarrow -5pe^t - 4qe^{2t} = 50e^t - 52e^{2t}$$

$$-5p = 50 \Rightarrow p = -10, \quad -4q = -52 \Rightarrow q = 13$$

$$\text{Particular Integral: } x = -10te^t + 13e^{2t}$$

$$\text{General Solution: } x = Ae^t + Be^{6t} - 10te^t + 13e^{2t}$$

$$x(0) = 0 \Rightarrow A + B = -13 \quad (1)$$

$$\dot{x} = Ae^t + 6Be^{6t} - 10te^t - 10e^t + 26e^{2t}$$

$$\dot{x}(0) = 1 \Rightarrow A + 6B = -15 \quad (2)$$

$$\text{Solve (1) and (2): } A = -\frac{63}{5}, \quad B = -\frac{2}{5}$$

$$\text{Particular Solution: } x = -\frac{63}{5}e^t - \frac{2}{5}e^{6t} - 10te^t + 13e^{2t}.$$

41 c)

$$\begin{aligned}
\text{Try this Particular Integral: } x &= pte^{6t} + qt + r \\
\dot{x} &= 6pte^{6t} + pe^{6t} + q, \quad \ddot{x} = 36pte^{6t} + 12pe^{6t} \\
\ddot{x} - 7\dot{x} + 6x &= 35e^{6t} - 36t \\
\Rightarrow 5pe^{6t} + 6qt + (-7q + 6r) &= 35e^{6t} - 36t \\
5p = 35 \Rightarrow p = 7, \quad 6q = -36 \Rightarrow q = -6, \quad 42 + 6r = 0 &\Rightarrow r = -7 \\
\text{Particular Integral: } x &= 7te^{6t} - 6t - 7 \\
\text{General Solution: } x &= Ae^t + Be^{6t} + 7te^{6t} - 6t - 7 \\
x(0) = 0 \Rightarrow A + B = 7 \quad (1) & \\
\dot{x} &= Ae^t + 6Be^{6t} + 42te^{6t} + 7e^{6t} - 6 \\
\dot{x}(0) = 1 \Rightarrow A + 6B = 0 \quad (2) & \\
\text{Solve (1) and (2): } A = \frac{42}{5}, \quad B &= -\frac{7}{5} \\
\text{Particular Solution: } x = \frac{42}{5}e^t - \frac{7}{5}e^{6t} &+ 7te^{6t} - 6t - 7.
\end{aligned}$$

41 d)

$$\begin{aligned}
\text{Try this Particular Integral: } x &= pt^2 + qt + r + m\cos 2t + n\sin 2t \\
\dot{x} &= 2pt + q - 2m\sin 2t + 2n\cos 2t \\
\ddot{x} &= 2p - 4m\cos 2t - 4n\sin 2t \\
\ddot{x} - 7\dot{x} + 6x &= 108t^2 - 100\cos 2t \\
\Rightarrow 6pt^2 + (-14p + 6q)t + (2p - 7q + 6r) & \\
+ (2m - 14n)\cos 2t + (14m + 2n)\sin 2t &= 108t^2 - 100\cos 2t \\
6p = 108 \Rightarrow p = 18, \quad -252 + 6q = 0 &\Rightarrow q = 42 \\
36 - 294 + 6r = 0 \Rightarrow r &= 43 \\
2m - 14n = -100 \quad (1), \quad 14m + 2n = 0 \quad (2) & \\
\text{Solve (1) and (2): } m = -1, \quad n &= 7 \\
\text{Particular Integral: } x &= 18t^2 + 42t + 43 - \cos 2t + 7\sin 2t \\
\text{General Solution: } x &= Ae^t + Be^{6t} + 18t^2 + 42t + 43 \\
&\quad - \cos 2t + 7\sin 2t \\
x(0) = 0 \Rightarrow A + B = -42 \quad (1) & \\
\dot{x} &= Ae^t + 6Be^{6t} + 36t + 42 + 2\sin 2t + 14\cos 2t \\
\dot{x}(0) = 1 \Rightarrow A + 6B = -55 \quad (2) & \\
\text{Solve (1) and (2): } A &= -\frac{197}{5} \\
B &= -\frac{13}{5} \\
\text{Particular Solution: } x &= -\frac{197}{5}e^t \\
&\quad - \frac{13}{5}e^{6t} + 18t^2 + 42t + 43 \\
&\quad - \cos 2t + 7\sin 2t.
\end{aligned}$$

42)

$$
\begin{aligned}
y'' + y &= \sin x + \sin 2x \\
\text{Homogeneous Equation: } y'' + y &= 0 \\
\text{Auxiliary Equation: } m^2 + 1 &= 0 \Rightarrow m^2 = -1 \Rightarrow m = \pm i \\
\text{Complementary Function: } y &= A\cos x + B\sin x \\
\text{Try this Particular Integral: } y &= px\cos x + qx\sin x \\
&\quad + m\cos 2x + n\sin 2x \\
y' &= -px\sin x + p\cos x + qx\cos x + q\sin x \\
&\quad - 2m\sin 2x + 2n\cos 2x \\
y'' &= -px\cos x - 2p\sin x - qx\sin x + 2q\cos x \\
&\quad - 4m\cos 2x - 4n\sin 2x \\
y'' + y &= \sin x + \sin 2x \\
\Rightarrow 2q\cos x - 2p\sin x - 3m\cos 2x - 3n\sin 2x &= \sin x + \sin 2x \\
2q = 0 \Rightarrow q = 0, \quad -2p = 1 \Rightarrow p &= -\frac{1}{2} \\
-3m = 0 \Rightarrow m = 0, \quad -3n = 1 \Rightarrow n &= -\frac{1}{3} \\
\text{Particular Integral: } y &= -\frac{1}{2}x\cos x - \frac{1}{3}\sin 2x \\
\text{General Solution: } y &= A\cos x + B\sin x \\
&\quad - \frac{1}{2}x\cos x - \frac{1}{3}\sin 2x \\
y(0) = 5 \Rightarrow A &= 5 \\
y &= 5\cos x + B\sin x \\
&\quad - \frac{1}{2}x\cos x - \frac{1}{3}\sin 2x \\
y' &= -5\sin x + B\cos x + \frac{1}{2}x\sin x \\
&\quad - \frac{1}{2}\cos x - \frac{2}{3}\cos 2x \\
y'(0) = 3 \Rightarrow 3 = B - \frac{1}{2} - \frac{2}{3} &\Rightarrow B = \frac{25}{6} \\
\text{Particular Solution: } y &= 5\cos x + \frac{25}{6}\sin x \\
&\quad - \frac{1}{2}x\cos x - \frac{1}{3}\sin 2x.
\end{aligned}
$$

43)

$$
\begin{aligned}
y'' + 2y' + 5y &= 17\cos 2x + 10\sin x \\
\text{Homogeneous Equation: } y'' + 2y' + 5y &= 0 \\
\text{Auxiliary Equation: } m^2 + 2m + 5 &= 0 \\
\Rightarrow (m+1)^2 - 1 + 5 = 0 &\Rightarrow (m+1)^2 = -4 \Rightarrow m = -1 \pm 2i \\
\text{Complementary Function: } y &= e^{-x}(A\cos 2x + B\sin 2x) \\
\text{Try this Particular Integral: } y &= p\cos 2x + q\sin 2x \\
&\quad + m\cos x + n\sin x \\
y' &= -2p\sin 2x + 2q\cos 2x - m\sin x + n\cos x \\
y'' &= -4p\cos 2x - 4q\sin 2x - m\cos x - n\sin x \\
y'' + 2y' + 5y &= 17\cos 2x + 10\sin x \\
\Rightarrow (p+4q)\cos 2x + (-4p+q)\sin 2x & \\
&\quad + (4m+2n)\cos x + (-2m+4n)\sin x = 17\cos 2x + 10\sin x \\
p + 4q &= 17\ (1), \quad -4p + q = 0\ (2) \\
\text{Solve (1) and (2): } p &= 1, \quad q = 4 \\
4m + 2n &= 0\ (3), \quad -2m + 4n = 10\ (4) \\
\text{Solve (3) and (4): } m &= -1, \quad n = 2 \\
\text{Particular Integral: } y &= \cos 2x + 4\sin 2x - \cos x + 2\sin x \\
\text{General Solution: } y &= e^{-x}(A\cos 2x + B\sin 2x) + \cos 2x + 4\sin 2x \\
&\quad - \cos x + 2\sin x \\
y(0) = 10 &\Rightarrow A = 10 \\
y &= e^{-x}(10\cos 2x + B\sin 2x) + \cos 2x + 4\sin 2x \\
&\quad - \cos x + 2\sin x \\
y' &= e^{-x}(-20\sin 2x + 2B\cos 2x) \\
&\quad - e^{-x}(10\cos 2x + B\sin 2x) \\
&\quad - 2\sin 2x + 8\cos 2x + \sin x + 2\cos x \\
y'(0) = 0 &\Rightarrow 2B - 10 + 8 + 2 = 0 \Rightarrow B = 0 \\
\text{Particular Solution: } y &= 10e^{-x}\cos 2x \\
&\quad + \cos 2x \\
&\quad + 4\sin 2x \\
&\quad - \cos x + 2\sin x.
\end{aligned}
$$